

PRODUCTION-READY ARCHITECTURE

EthioScore Engine: Alternative Credit Risk Infrastructure & Comprehensive Technical Blueprint

An Aligned Framework Bridging Credit Exclusions, Modern Machine-Aided Risk Signals, and National Bank of Ethiopia Directive SBB/80/2022.

Architect & System Designer: Natnael Seyum

Credentials: Management & Veterinary Pharmacy Dual Discipline Focus

Contact E-mail: natnaelseyum20@gmail.com

Contact Phone: +251 915 732 652

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1. Executive Summary & Vision Block

Traditional credit underwriting frameworks in emerging economies remain fundamentally dependent on physical real estate collateral and mature formal income histories. In the context of the Ethiopian credit landscape, this reliance produces a profound **"thin-file" problem**, rendering millions of smallholder agriculturalists, micro-merchants, and youth entrepreneurs systematically invisible to commercial bank books.

EthioScore Engine resolves this structural asymmetry. By deploying a calibrated risk calculation layer built using alternative data pipelines, the application demonstrates how digital telecom transactions, regional structural variances, and localized cooperative ecosystem footprints can safely extend banking books while strictly validating macro-prudential regulations. It serves as a self-contained proof-of-concept for next-generation automated underwriting.

Core Paradigm Shift

Moving from traditional fixed collateral requirements to dynamic transaction-based behavior profiles allows institutions to capture unbanked credit signals cleanly while enforcing automated compliance mapping.

2. Interface Deep-Dive: Core Modules & Architecture

A. Dashboard Home: System Foundations

The platform opens with a structured macro-overview, introducing the baseline metrics driving the system. It builds consumer and regulatory trust upfront by revealing the model parameters natively:

- **Standardized Scoring Array:** Operates across a transparent **300 to 850 FICO-style index range** for immediate industry familiarity.
- **Alternative Signal Multipliers:** Quantifies alternative transaction matrices at approximately **37% of the total available score weight**, proving that data points outside legacy payrolls can safely drive core credit outcomes.
- **Integrated Classification Architecture:** Sets up **5 NBE credit bands** directly within the onboarding process.

B. Live Credit Score Simulator

The Live Scorer acts as an interactive playground demonstrating real-time model mechanics. It accepts a blend of standard and high-impact alternative data dimensions, visualizing their joint effect on risk assessment instantly:

- **Dynamic Mathematical Scoring Yield:** When inputting a baseline profile (e.g., 8,500 ETB income, 28% DTI, 14 months mobile money consistency, and a poor repayment record), the engine calculates a precise score of **573 ("Fair")**.
- **Automated Risk Actions:** Because a 573 score reflects borderline stability, the engine automatically routes the file to **MANUAL REVIEW**, capping maximum loan exposure to **25,000 ETB** with an expected loss provisioning rate of **ETB 900** (matching the mandated 3% provisioning rule).
- **Granular Factor Attribution:** The UI displays an explicit *Score Factor Breakdown*, visualizing feature weight metrics (Income Level at 100%, Debt Burden at 72%, Mobile Money at 69%, Asset Base at 67%, and Repayment History at 0%). This achieves complete model transparency.

3. Macro Portfolio & Pipeline Analytics

The **Applicant Pipeline** panel functions as the institutional credit command hub, validating model performance over a scaled cohort of 300 programmatically simulated individuals.

Macro Portfolio Performance Metric	Observed Diagnostic Value	Institutional Risk Implication
Total Applications	300 Profiles	Proves pipeline resilience and database integration capacity.
Average Credit Portfolio Score	596 (Fair Classification)	Validates dataset integrity; mirrors a true unbanked distribution curve.
Automated Approval Rate	22.3%	Reflects high risk-aversion, maintaining tight capital controls.
Total Approved Asset Book	ETB 2,483,000	Demonstrates market-scaling capability without exposing banks to excessive default.

The application couples these portfolio analytics with a regional scoring distribution chart, keeping credit officers aware of geographic exposure and concentration risks across administrative zones (e.g., Addis Ababa, Dire Dawa, Harari, Amhara, SNNPR).

4. Regulatory Integration & Fairness Audit

A. Strict Compliance with NBE Directive SBB/80/2022

Rather than keeping regulatory compliance as an afterthought, the EthioScore Engine natively codes national underwriting law into its technical routing matrix:

- **Art. 5 & 6 (Asset Classification & Provisioning):** Automated classification logic ties individual scores directly to legal designations—mapping records instantly from *Pass (1% provision)* up to *Doubtful/Loss (50% provision)* boundaries.
- **Art. 9 (Sector Concentration Alerts):** Active flags evaluate portfolio exposure across diverse sectors (Trade/SME, Agriculture, Manufacturing, Civil Servants) to avoid single-sector lending shocks.
- **Identified Infrastructure Gaps:** The platform highlights institutional integration milestones, explicitly listing collateral tracking (Art. 8) and cross-border data residency boundaries as manual items to resolve during core legacy banking deployments.

B. Algorithmic Fairness & Score Parity Audit

Emerging systems must guarantee equity across demographic splits. EthioScore features an embedded **Fairness Audit matrix** tracking data performance across Urban/Rural geographies and Age cohorts:

Geographic Tier	Demographic Segment	Sample Size (n)	Mean Credit Score	Approval Rate (%)
Rural	Youth (<30)	55	566.7	10.9%
Rural	Prime (30–44)	62	595.0	25.8%
Rural	Senior (45+)	91	585.5	14.3%
Urban	Youth (<30)	25	630.7	36.0%
Urban	Prime (30–44)	28	612.1	32.1%
Urban	Senior (45+)	39	627.7	35.9%

System Optimization Insight

Audit Flag Detected: Youth applicants under 30 show a systematically lower approval rate (10.9% in rural sectors). **Prescribed Remediation:** Adjust model hyperparameters to dynamically re-weight mobile money velocity indicators and cooperative history, balancing out shorter traditional employment lifespans without increasing risk exposure.

5. Technical Stack & Production Roadmap

The prototype is fully containerized and operational via a high-performance Python architecture utilizing **Streamlit** for front-end reactive state management, alongside **Pandas** and **NumPy** for vector computation and pipeline processing. To scale this prototype to a national market grade, the infrastructure roadmap defines three main deployment milestones:

- 1. Production API Integration:** Replacing the stochastic simulation loop with secure, encrypted REST webhooks connecting straight to telecom utility endpoints (Telebirr/CBE Birr core data layers) under a strict user consent framework.
- 2. National Identity Handshake:** Merging user records against Ethiopia's centralized **Fayda ID system** to automate KYC validation and prevent identity fraud across lending pools.
- 3. Relational Persistence:** Shifting memory states to a secure relational database (e.g., PostgreSQL) with native encryption protocols, establishing an ironclad audit trail for National Bank of Ethiopia oversight panels.